The complete equidistribution classification of length-three mesh patterns on layered permutations

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Abstract. The distributional classification of short mesh patterns has developed through a sequence of results for selected pattern families and host classes. A recent complete classification at length two on layered permutations gives 35 distribution classes, whereas a length-three mesh pattern may be chosen in $6 \cdot 2^{16} = 393{,}216$ ways and complete distributional results have previously been out of reach. We show that a full classification is possible on layered permutations. For mesh patterns with layered underlying permutation, we introduce a layer-constraint signature and prove occurrence and signature-enumeration formulas valid at arbitrary length; at length three these tools yield exactly 147 distribution-equivalence classes, including the zero class. All equidistribution identities used in the classification are proved uniformly for arbitrary host size. To the best of our knowledge, this is the first complete distributional classification of length-three mesh patterns on any nontrivial host class.

Keywords: mesh pattern, layered permutation, equidistribution, composition, classification.

AMS Subject Classifications: 05A05, 05A15.

1 Introduction

Let S_n denote the set of permutations of $[n] = \{1, \ldots, n\}$. If $\pi = \pi_1 \cdots \pi_n \in S_n$ and $\tau = \tau_1 \cdots \tau_k \in S_k$, an *occurrence* of the classical pattern τ in π is a sequence of indices $1 \le i_1 < \cdots < i_k \le n$ such that

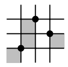
$$\pi_{i_a} < \pi_{i_b} \iff \tau_a < \tau_b \qquad (1 \le a, b \le k).$$

Thus the selected entries have the same relative order as τ. For example, the subsequence $\pi_1\pi_2\pi_4 = 254$ of $\pi = 25143$ is an occurrence of 132. We refer to Kitaev [7] for general background on classical permutation patterns.

Mesh patterns, introduced by Brändén and Claesson [3], refine this notion by imposing emptiness conditions around a classical occurrence. A mesh pattern of length k is a pair $p = (\tau, R)$, where $\tau \in S_k$ and $R \subseteq [0, k] \times [0, k]$, with $[0, k] = \{0, 1, \ldots, k\}$. Fix an occurrence at positions $i_1 < \cdots < i_k$, let $v_1 < \cdots < v_k$ be its selected values in increasing order, and set

$$i_0 = v_0 = 0, \qquad i_{k+1} = v_{k+1} = n + 1.$$

It is an occurrence of p when, for every $(a, b) \in R$, the open rectangle $(i_a, i_{a+1}) \times (v_b, v_{b+1})$ contains no point (j, π_j). In the standard diagram one plots the points (a, τ_a) and shades the unit cells indexed by R. For example,



$$p = \big(132, \{(0,0), (1,1), (1,2), (3,1)\}\big).$$

The entries 243 in the permutation 1243 form a classical occurrence of 132, but not an occurrence of the displayed mesh pattern: the unselected point $(1, 1)$ lies in the shaded cell $(0, 0)$. By contrast, the first three entries of 1324 do form an occurrence, since its only unselected point lies in the unshaded cell $(3, 3)$. To save space, throughout the paper we write the underlying permutation in one-line notation and record the shaded set algebraically; after the layer reduction we use the still more compact signature notation.

Write $p(\pi)$ for the number of occurrences of p in π. For a fixed host family $\mathcal{C}$, define

$$D^{\mathcal{C}}_{p,n}(q) = \sum_{\substack{\pi \in \mathcal{C} \\ |\pi| = n}} q^{p(\pi)}.$$

Two patterns are *distribution equivalent* on $\mathcal{C}$ when these polynomials agree for every n. This is distinct from coincidence, which asks for equality of the avoidance sets, and from Wilf-equivalence, which asks only for equality of their cardinalities. Because a mesh pattern of length k has $(k+1)^2$ cells, there are $k!\, 2^{(k+1)^2}$ raw patterns. Our problem is to classify the $6 \cdot 2^{16} = 393{,}216$ patterns of length three by distribution equivalence when $\mathcal{C}$ is the class of layered permutations.

The literature on short mesh patterns began with avoidance. Hilmarsson *et al.* [6] initiated the Wilf-classification at length two, reducing the 1,024 raw patterns to at most 65 candidates and then to an upper bound of 56 Wilf-classes. Kitaev and Zhang [9] began the systematic study of full occurrence distributions for these patterns. Further length-two equidistributions and infinite-family extensions were obtained by Han and Zeng [5] and by Kitaev, Zhang, and Zhang [10]. Related short-pattern statistics have also been pursued on restricted hosts, including alternating permutations [8, 4], while distributions of selected ordinary length-two mesh patterns have been studied on king permutations [11].

At length three, Bean, Gudmundsson, Magnusson, and Ulfarsson [2] completed the coincidence classification. That result determines when two patterns have exactly the same avoiders, but it does not determine their full occurrence distributions. Distributional work at length three has instead concentrated on structured subfamilies, notably joint equidistributions of mesh patterns with underlying permutations 123 and 132 or 123 and 321 and with symmetric or minus-antipodal shadings [12, 13, 14]. Thus, despite substantial progress, these studies do not classify the distributions of all 393,216 length-three mesh patterns.

Layered permutations provide a setting in which a complete result becomes possible. They form a natural and well-studied permutation class [1]; every layered permutation is encoded by a composition, and every classical occurrence has a canonical description in terms of supporting layers. Our recent length-two classification on this class [16] used that description case by case and obtained 35 distribution classes. To the best of our knowledge, the present paper is the first to show that the full distributional classification of length-three mesh patterns can also be completed on a specified nontrivial host class.

Our main tool is a layer-constraint signature that we introduce. The active-cell theorem and the signature formula are valid for mesh patterns of arbitrary length whose underlying permutation is layered. They yield a uniform rational generating function for total occurrences and an explicit enumeration of signatures at every length. At length three, they reduce the 393,216 raw diagrams first to 1,600 effective shadings and then to 644 signatures. Exact pointwise identities produce 523 pointwise classes, reverse-complement gives 286 pointwise–rc orbits, the kernel-degree argument reduces these to 200, and layer-coordinate relabelling gives 164. Cut shifts, non-overlapping contractions, separator-word complementation, one exceptional rational identity, and a major-index–inversion identity complete the remaining identifications. The successive reductions have sizes

$$644 \longrightarrow 523 \longrightarrow 286 \longrightarrow 200 \longrightarrow 164 \longrightarrow 146.$$

There are 146 nonzero distribution classes, and the two non-layered underlying permutations contribute one additional zero class, for a total of 147. The upper bound is established by all-size identities, and the lower bound by exact separation of the resulting 146 rows. Figure 1 summarizes the two levels of the argument.

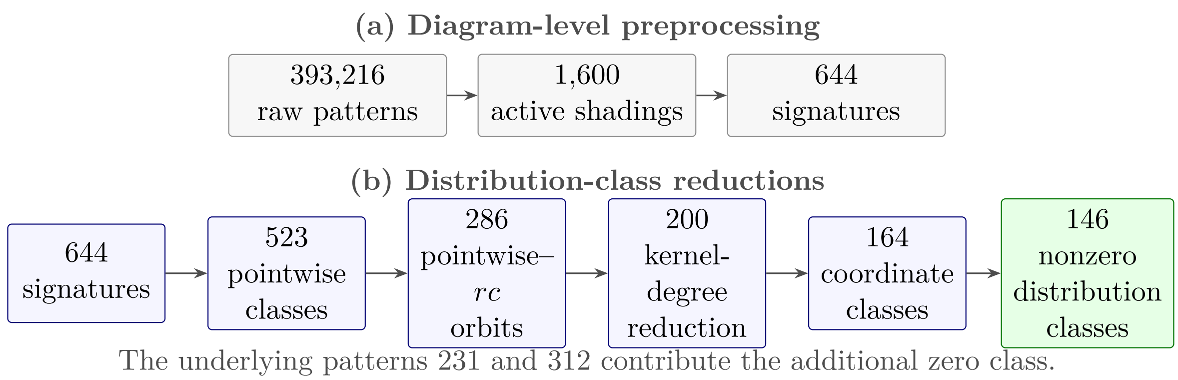


Figure 1: The two levels of the classification. Panel (a) records the diagram-level preprocessing. Panel (b) gives the distribution-class reductions; its arrows represent, from left to right, pointwise equality, reverse-complement, kernel-degree reduction, layer-coordinate relabelling, and the remaining distributional bridges.

The rest of the paper develops this outline without changing levels of equivalence implicitly. Section 2 identifies the active cells on layered hosts. Section 3 derives the signature formula, enumerative consequences, and the exact pointwise quotient. Section 4 proves the all-size distributional reductions, and Section 5 assembles the complete classification. The appendix proves the uniform coordinate identities. The accompanying short Supplementary Proof Details record only the finite incidence data needed to audit the last two counts: the 43 coordinate connections, the inventory of the eighteen bridges, and the pairwise separation statement. All identities valid for arbitrary host size are proved in the main paper.

2 Layered hosts and active cells

A permutation is *layered* if it is a direct sum of decreasing permutations. For positive integers $\ell_1, \dots, \ell_m$, write

$$L(\ell_1, \dots, \ell_m) = \delta_{\ell_1} \oplus \cdots \oplus \delta_{\ell_m}, \qquad \delta_r = r(r-1)\cdots 1.$$

The class of all layered permutations is denoted by $\mathcal{L}$. Its members of size n are in bijection with the compositions of n. We henceforth suppress the superscript $\mathcal{L}$ in the distribution polynomial $D^{\mathcal{L}}_{p,n}(q)$.

Every subpermutation of a layered permutation is layered. The layered permutations of length three are

$$123 = \delta_1 \oplus \delta_1 \oplus \delta_1, \quad 132 = \delta_1 \oplus \delta_2, \quad 213 = \delta_2 \oplus \delta_1, \quad 321 = \delta_3.$$

Consequently, the underlying permutations 231 and 312 give the zero statistic, regardless of the shading.

Suppose that the layered underlying permutation has block sizes $r = (r_1, \dots, r_s)$, and put $R_j = r_1 + \cdots + r_j$, with $R_0 = 0$. Define the internal channel of block j and the external boundary cells by

$$\mathcal{C}_j = \{(R_{j-1} + h, R_j - h) : 0 \le h \le r_j\}, \qquad \mathcal{E}_j = (R_j, R_j) \quad (0 \le j \le s).$$

Theorem 2.1 (Active cells)**.** *For mesh patterns with layered underlying permutation $\delta_{r_1} \oplus \cdots \oplus \delta_{r_s}$, the cells that can contain an unselected point relative to an occurrence in a layered host are exactly*

$$\mathcal{C}_1 \cup \cdots \cup \mathcal{C}_s \cup \{\mathcal{E}_0, \dots, \mathcal{E}_s\}.$$

Shading any other cell has no effect on the occurrence statistic on $\mathcal{L}$.

Proof. Every selected decreasing block lies in one host layer. An unselected point in a supporting layer lies in the corresponding internal channel, because position and value ranks change in opposite directions inside a decreasing layer. An unselected layer before, between, or after the supporting layers lies in the corresponding diagonal boundary cell. This proves that no other cell can be occupied. Conversely, each channel cell is realized by inserting an extra point in the appropriate internal gap, and each boundary cell is realized by inserting an unused host layer in the corresponding external gap. Hence every listed cell is active. □

There are $3 + 2s + 1$ active cells at length three. Figure 2 displays their two types for each layered underlying permutation, and Table 1 records the resulting counts.

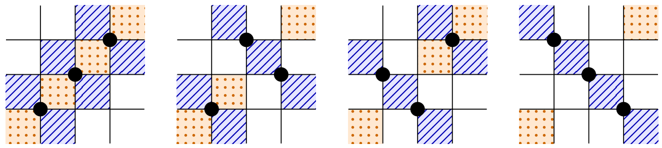


Figure 2: Active cells for the four layered permutations of length three. Blue diagonally hatched cells are internal-channel cells, and orange dotted cells are external boundary cells; unmarked cells are inactive. These markings indicate activity, not mesh shading.

Table 1: Active-cell and signature counts at length three.

τ	block sizes	active cells	effective shadings	signatures
123	$(1,1,1)$	10	$2^{10} = 1{,}024$	$3^3 2^4 = 432$
132	$(1,2)$	8	$2^8 = 256$	$(3 \cdot 4)2^3 = 96$
213	$(2,1)$	8	$2^8 = 256$	$(4 \cdot 3)2^3 = 96$
321	(3)	6	$2^6 = 64$	$5 \cdot 2^2 = 20$

Thus the four occurring underlying permutations have 1,600 effective shadings. All inactive-cell choices are restored at the end by multiplying with the appropriate power of two.

3 Signatures, enumeration, and pointwise statistics

Let z_j be the number of shaded cells in $\mathcal{C}_j$, and let $E \subseteq \{0, 1, \dots, s\}$ record the shaded external cells. The layer-constraint signature is

$$\Sigma = (r, z, E) = ((r_1, \dots, r_s), (z_1, \dots, z_s), E).$$

For a supporting layer of size ℓ, define

$$K_{r,z}(\ell) = \begin{cases} \dbinom{\ell - z}{r - z}, & 0 \le z \le r, \\ \mathbf{1}_{\ell = r}, & z = r + 1. \end{cases}$$

Here and below, $\binom{a}{b} = 0$ unless $a \ge b \ge 0$; in particular, $K_{r,z}(\ell) = 0$ when $\ell < r$. The first case follows by distributing the $\ell - r$ unselected points among the $r + 1 - z$ unshaded internal gaps. The second case is rigid: all internal gaps must be empty.

For a host composition $L = (\ell_1, \dots, \ell_m)$, let $\Phi_E(m)$ consist of the increasing maps from $\{1, \dots, s\}$ to $\{1, \dots, m\}$ satisfying

$$0 \in E \Rightarrow \phi(1) = 1, \qquad s \in E \Rightarrow \phi(s) = m, \qquad j \in E \cap \{1, \dots, s-1\} \Rightarrow \phi(j+1) = \phi(j)+1.$$

Thus the elements 0 and s encode only the two endpoint conditions, while the internal elements of E encode adjacency. We set $\Phi_E(m) = \varnothing$ when $m < s$; when $s = 1$ there is no internal adjacency condition.

Theorem 3.1 (Signature formula)**.** *For every layered host $L = (\ell_1, \dots, \ell_m)$,*

$$P_\Sigma(L) = \sum_{\phi \in \Phi_E(m)} \prod_{j=1}^{s} K_{r_j,z_j}(\ell_{\phi(j)}).$$

Consequently, equal signatures determine equal statistics on every individual layered permutation, so the distributional classification on layered hosts factors through signatures.

Proof. Choose the supporting host layers through ϕ. The shaded external cells give precisely the boundary and adjacency conditions defining $\Phi_E(m)$. Once the supporting layers are fixed, the choices inside distinct layers are independent, and the jth layer contributes K_{r_j,z_j}. Multiplication counts the occurrences with that support, and summation over the admissible supports gives the formula. □

For later use, let

$$\Lambda(t) = 1 + \sum_{n \geq 1} 2^{n-1} t^n = \frac{1-t}{1-2t},$$

the ordinary generating function for compositions, including the empty composition. For a signature $\Sigma = (r, z, E)$, put $f(\Sigma) = s + 1 - |E|$ and

$$\alpha(\Sigma) = \sum_{\substack{1 \leq j \leq s \\ z_j \leq r_j}} (r_j - z_j + 1).$$

Proposition 3.2 (Total-occurrence generating function at arbitrary length)**.** *For every signature Σ of length k,*

$$T_\Sigma(t) := \sum_{\pi \in \mathcal{L}} P_\Sigma(\pi) t^{|\pi|} = \frac{t^k \Lambda(t)^{f(\Sigma)}}{(1-t)^{\alpha(\Sigma)}}.$$

Proof. Fix an occurrence of Σ. Each unshaded external gap may contain an arbitrary, possibly empty, sequence of unused host layers, and therefore contributes the composition series $\Lambda(t)$. Hence the $f(\Sigma)$ free external gaps contribute $\Lambda(t)^{f(\Sigma)}$.

For a non-rigid supporting coordinate,

$$\sum_{\ell \geq 1} K_{r,z}(\ell) t^\ell = \frac{t^r}{(1-t)^{r-z+1}},$$

whereas a rigid coordinate contributes t^r. Multiplying these contributions over the supporting layers and using $\sum_j r_j = k$ gives the stated formula. □

Proposition 3.3 (Signature enumeration at arbitrary length)**.** *Let S_k be the number of layer-constraint signatures arising from length-k mesh patterns with layered underlying permutation. Then*

$$S_k = \sum_{(r_1,\dots,r_s)\models k} 2^{s+1}\prod_{j=1}^{s}(r_j+2), \qquad \sum_{k\geq 1} S_k x^k = \frac{12x-8x^2}{1-8x+5x^2}.$$

Equivalently, $S_1 = 12$, $S_2 = 88$, and $S_k = 8S_{k-1} - 5S_{k-2}$ for $k \geq 3$; in particular, $S_3 = 644$ and $S_4 = 4{,}712$.

Proof. For a fixed block composition $(r_1,\dots,r_s)$, each z_j has r_j+2 possible values and $E \subseteq \{0,\dots,s\}$ has 2^{s+1} possible values. This gives the stated sum. Put

$$A(x) = \sum_{r\geq 1} 2(r+2)x^r = \frac{2x(3-2x)}{(1-x)^2}.$$

Since $2^{s+1}\prod_{j=1}^{s}(r_j+2) = 2\prod_{j=1}^{s}\big(2(r_j+2)\big)$, summing over all nonempty block compositions gives

$$\sum_{k\geq 1} S_k x^k = 2\sum_{s\geq 1} A(x)^s = \frac{2A(x)}{1-A(x)} = \frac{12x-8x^2}{1-8x+5x^2}.$$

Comparing coefficients gives the stated recurrence and initial values. $\square$

The number of effective mesh patterns represented by a signature is $\nu(\Sigma) = \prod_{j=1}^{s}\binom{r_j+1}{z_j}$. At length three, summing these multiplicities over the $S_3 = 644$ signatures restores the 1,600 effective shadings. Every subsequent classification step is carried out on the 644 signatures.

Equal signatures are not necessary for pointwise equality. Put

$$\mathcal{B} = \left(1, x, \tbinom{x}{2}, \tbinom{x}{3}, \mathbf{1}_{x=1}, \mathbf{1}_{x=2}, \mathbf{1}_{x=3}\right).$$

Every length-three kernel $K_{r,z}$ has an exact expansion in $\mathcal{B}$. The cutoff functions are important; for example, $K_{2,2}(x) = 1 - \mathbf{1}_{x=1}$ rather than the constant function one.

The seven functions in $\mathcal{B}$ are linearly independent on $\mathbb{Z}_{>0}$. Indeed, a linear relation restricted to $x \geq 4$ forces its polynomial part, of degree at most three, to vanish identically; evaluating the remaining relation at $x = 1, 2, 3$ then forces the three indicator coefficients to vanish. It follows, successively in each coordinate, that $\mathcal{B}^{\otimes m}$ is linearly independent for every m. Hence, for every number m of host layers, the tensor expansion is unique.

Theorem 3.4 (Exact pointwise quotient)**.** *The* 644 *signatures induce exactly* 523 *distinct occurrence statistics on individual layered permutations.*

Proof. Let X denote the formal letter corresponding to the constant function 1 in $\mathcal{B}$, and put $G = 1 + X + X^2 + \cdots = (1-X)^{-1}$. Expand each kernel $K_{r,z}$ as the corresponding linear form $\kappa_{r,z}$ in the seven letters of $\mathcal{B}$. The full family of coefficient tensors of a signature is the noncommutative formal series

$$G_0\kappa_{r_1,z_1}G_1\cdots\kappa_{r_s,z_s}G_s, \qquad G_j=\begin{cases}1, & j\in E,\\ G, & j\notin E.\end{cases}$$

Indeed, G_j records the unused host layers in the jth external gap.

This expression has a finite canonical form. Expand every $\kappa_{r,z}$ into its X-part and its six remaining letters, and collect terms according to the ordered word of non-X letters. Between two consecutive such letters there remains only a rational series in X, obtained from at most four factors chosen from 1 and G and at most three explicit factors X. For a fixed word of non-X letters, introduce one variable for each intervening run and put every denominator over $(1-X_i)^4$; the resulting polynomial numerator is unique. Thus equality is decided by comparing a finite list of integer numerator polynomials. Performing this coefficientwise reduction on the 644 signatures gives 523 distinct canonical forms. Equal forms have identical tensors for every m, and distinct forms differ as formal series; the independence of $\mathcal{B}^{\otimes m}$ then proves the assertion. Thus the quotient is obtained from exact formal-series comparison, not from agreement on a bounded collection of hosts. □

Remark 3.5. *The number* 523 *results from collecting the* 644 *canonical forms described in the proof. Since the criterion is an identity of formal series, this step does not infer equality from agreement on hosts with a bounded number of layers.*

4 From signatures to distributions

4.1 Reverse-complement after the pointwise quotient

Reverse-complement rotates a permutation plot by 180 degrees and preserves the class $\mathcal{L}$. At the signature level it acts by

$$(r_1,\dots,r_s;z_1,\dots,z_s;E)\longmapsto(r_s,\dots,r_1;z_s,\dots,z_1;s-E),$$

where $s-E=\{s-e:e\in E\}$. Inversion fixes the signature: it only reverses the cells inside each internal channel and therefore preserves their number z_j, while every external cell lies on the diagonal. Thus inversion gives no additional quotient after signatures have been formed.

The exact pointwise relation is preserved by reverse-complement, so this involution descends from signatures to the 523 pointwise classes of Theorem 3.4.

Proposition 4.1 (Pointwise reverse-complement quotient)**.** *The* 523 *pointwise classes form* 286 *pointwise–rc orbits.*

Proof. Exactly 49 pointwise classes are fixed by the induced involution, while the remaining 474 form 237 pairs. Burnside's lemma therefore gives $49+474/2=286$. The fixed-class count is taken directly on the canonical formal-series representatives of Theorem 3.4. □

4.2 Complete kernel-degree reduction

Let $f(\Sigma) = s + 1 - |E|$ be the number of free external gaps. The condition $f \leq 1$ used below is a genuine restriction, not an automatic consequence of working at length three. Indeed, the possible values are

underlying pattern	s	possible values of f
123	3	$0, 1, 2, 3, 4$
$132, 213$	2	$0, 1, 2, 3$
321	1	$0, 1, 2.$

When $f \leq 1$, all unused host layers must lie in the unique free external gap, if such a gap exists. Consequently every host admitting the prescribed core has a unique support injection. If $f \geq 2$, by contrast, unused layers can be distributed among several free gaps, and the occurrence statistic becomes a sum over several possible supports. The resulting global support geometry is not captured by the unordered local data below; these multiple-support cases are treated in the next two subsections.

For the unique-support range, write a supporting layer as $\ell_j = r_j + u_j$. For every non-rigid coordinate put $d_j = r_j - z_j$ and define the degree multiset $D(\Sigma) = \{\{d_j : z_j \leq r_j\}\}$. Rigid coordinates are omitted because they force $u_j = 0$ and contribute the factor one. The local occurrence number becomes

$$\omega_D(\mathbf{u}) = \prod_{d_j \in D} \binom{u_j + d_j}{d_j}.$$

For $f(\Sigma) \leq 1$, define

$$F_\Sigma(t, q) = \sum_{\pi \in \mathcal{L}} q^{P_\Sigma(\pi)} t^{|\pi|},$$
$$C_D(t, q) = t^3 \sum_{\mathbf{u} \geq 0} t^{|\mathbf{u}|} \big(q^{\omega_D(\mathbf{u})} - 1\big).$$

Proposition 4.2 (Distribution series in the unique-support range)**.** *If $f(\Sigma) \leq 1$, then*

$$F_\Sigma(t, q) = \begin{cases} \Lambda(t) + C_D(t, q), & f(\Sigma) = 0, \\ \Lambda(t)\big(1 + C_D(t, q)\big), & f(\Sigma) = 1. \end{cases}$$

Proof. Start with the contribution $\Lambda(t)$ of all layered hosts, each carrying weight one. A host with a nonzero occurrence contains the prescribed core with a unique support. Replacing its baseline weight 1 by $q^{\omega_D(\mathbf{u})}$ gives the correction $C_D(t, q)$ when there is no free external gap. When there is one free external gap, that gap contains an arbitrary composition and contributes the additional factor $\Lambda(t)$. □

Corollary 4.3 (Avoidance in the unique-support range)**.** *Let*

$$A_\Sigma(t) = \sum_{\substack{\pi \in \mathcal{L} \\ P_\Sigma(\pi) = 0}} t^{|\pi|}.$$

If $f(\Sigma) \le 1$, then

$$A_\Sigma(t) = \Lambda(t) - \frac{t^3 \Lambda(t)^{f(\Sigma)}}{(1-t)^{|D(\Sigma)|}}.$$

Consequently, in the unique-support range the avoidance generating function depends only on $f(\Sigma)$ and $|D(\Sigma)|$.

Theorem 4.4 (Complete unique-support classification)**.** *Let Σ and Σ' be length-three signatures with at most one free external gap. Then they are equidistributed on $\mathcal{L}$ if and only if*

$$f(\Sigma) = f(\Sigma') \qquad \text{and} \qquad D(\Sigma) = D(\Sigma').$$

Proof. For sufficiency, Proposition 4.2 shows directly that the distribution series depends only on (f, D).

For necessity, suppose that Σ and Σ' are equidistributed. By Proposition 3.2, their total-occurrence series are equal. Since $f \in \{0, 1\}$, the pole at $t = 1/2$ determines f, and hence $f(\Sigma) = f(\Sigma')$.

After removing the common free-gap factor, Proposition 4.2 implies that the corresponding correction series $C_D(t, q)$ agree. The universal initial term is $t^3(q-1)$. The first subsequent coefficient that depends on D is

$$[t^4]C_D(t,q) = \sum_{d \in D} (q^{d+1} - 1),$$

because exactly one surplus variable equals one. This polynomial records every element of D, including its multiplicity. Hence $D(\Sigma) = D(\Sigma')$. □

Remark 4.5 (Why the unique-support hypothesis is sharp)**.** *Consider*

$$\Sigma_1 = (111; 000; \{0,1\}), \qquad \Sigma_2 = (111; 000; \{0,2\}).$$

Both signatures have $f = 2$ and $D = \{\{1,1,1\}\}$, but their support geometries are different. On the host composition $L = (1,1,2,2)$, the first signature permits the third support to be either Layer 3 or Layer 4, whereas the second has admissible supports $(1,2,3)$ and $(1,3,4)$. Therefore

$$P_{\Sigma_1}(L) = \ell_1 \ell_2 (\ell_3 + \ell_4) = 4,$$
$$P_{\Sigma_2}(L) = \ell_1 \ell_2 \ell_3 + \ell_1 \ell_3 \ell_4 = 6.$$

Thus

$$D_{\Sigma_1,6}(q) = 6 + 10q^4 + 14q^6 + 2q^8, \qquad D_{\Sigma_2,6}(q) = 6 + 9q^4 + 2q^5 + 13q^6 + 2q^8.$$

Hence the two signatures are not equidistributed. This is a concrete instance of the lower-bound separation used in Theorem 5.1.

At length three the possible degree multisets are

$$\varnothing, \quad \{0\}, \{1\}, \{2\}, \{3\}, \quad \{0,0\}, \{0,1\}, \{0,2\}, \{1,1\}, \{1,2\},$$
$$\{0,0,0\}, \{0,0,1\}, \{0,1,1\}, \{1,1,1\}.$$

Consequently, the 56 fully constrained signatures form exactly fourteen distribution classes, and the 190 signatures with one free gap form another fourteen distribution classes. Thus 246 signatures are classified into 28 distribution classes. In the pointwise–rc quotient this replaces 114 pointwise–rc orbits by 28 distribution classes and improves the bound 286 to

$$286 - 114 + 28 = 200.$$

For example, the three local data

$$(r, z) = (3, 2), \qquad (r, z) = ((1, 2), (0, 3)), \qquad (r, z) = ((1, 1, 1), (0, 2, 2))$$

all have degree multiset $D = \{1\}$, despite arising from different underlying layered permutations.

4.3 Layer-coordinate relabelling

Theorem 4.4 completes the unique-support range $f \leq 1$. We now consider the pointwise–rc orbits represented by the remaining signatures with $f \geq 2$. In these multiple-support cases a host may admit several support injections, so $P_\Sigma(L)$ is a sum of kernel products, and the unordered degree multiset $D(\Sigma)$ no longer records how the summands are positioned among the host layers.

Some of these sums nevertheless agree after a permutation of the variables $\ell_1, \ldots, \ell_m$. Such a relabelling acts on the host composition $L = (\ell_1, \ldots, \ell_m)$, not on the mesh pattern. For fixed n and m it is a size-preserving bijection on the compositions of n with m parts, and therefore gives the following composition-level criterion.

Theorem 4.6 (Layer-coordinate criterion)**.** *Suppose that for every m there is $\rho_m \in S_m$ such that*

$$P_\Sigma(\ell_1, \ldots, \ell_m) = P_{\Sigma'}(\ell_{\rho_m(1)}, \ldots, \ell_{\rho_m(m)})$$

for all positive ℓ_i. Then Σ and Σ' are equidistributed on $\mathcal{L}$.

Proof. The coordinate permutation is a bijection of the compositions of each fixed n and m. The displayed identity therefore gives the same multiset of occurrence values. Summing over m proves equality of the distribution polynomials for every n. □

Unlike Theorem 4.4, the layer-coordinate criterion is only sufficient: it does not assert that every multiple-support equidistribution arises from a coordinate permutation. It supplies uniform bijections for the identifications recorded below, while the cases not reached by such permutations require the composition bijections of Section 4.4.

For example, put $\Sigma_A = (111; 000; \{0, 1\})$ and $\Sigma_B = (111; 000; \{0, 3\})$. For Σ_A, the constraints $0, 1 \in E$ force the first two supporting layers to be Layers 1 and 2, while the third support may be any later layer. For Σ_B, the first and third supports are Layers 1 and m, while the middle support may be any intermediate layer. Since every local kernel is $K_{1,0}(\ell) = \ell$, the two occurrence polynomials are

$$P_{\Sigma_A}(L) = \ell_1 \ell_2 \sum_{j=3}^{m} \ell_j \qquad \text{and} \qquad P_{\Sigma_B}(L) = \ell_1 \ell_m \sum_{j=2}^{m-1} \ell_j.$$

Under the coordinate bijection

$$L = (\ell_1, \ell_2, \ell_3, \dots, \ell_m) \longmapsto L' = (\ell_1, \ell_3, \dots, \ell_m, \ell_2),$$

we have $P_{\Sigma_B}(L') = \ell_1 \ell_2 \sum_{j=3}^{m} \ell_j = P_{\Sigma_A}(L)$. Thus the two statistics are equidistributed even though their support geometries differ and the signatures are not related by reverse-complement.

The uniform coordinate permutations are listed in Appendix A. There Lemma A.1 proves their identities for arbitrary m.

After the kernel-degree reduction, 172 multiple-support pointwise–*rc* orbits remain. Of these, 84 are not incident with a coordinate edge. The 43 coordinate edges on the other 88 orbits have 36 effective unions and leave 52 components. Supplementary Table S2 gives the source signature, target signature, orientation, and coordinate word for every edge. The resulting upper bound is therefore

$$28 + 84 + 52 = 164.$$

4.4 The remaining distributional bridges

After the kernel-degree reduction and layer-coordinate relabelling, the local kernel data and the coordinate-level support symmetries have been exhausted. The remaining candidate identifications among the 164 coordinate classes must therefore be sought at the level of the host compositions themselves. In these residual cases, the relevant differences arise from the placement of cuts and marked adjacent blocks in the composition, rather than from any further symmetry of the mesh diagrams.

Every layered host is represented by a composition $L = (\ell_1, \dots, \ell_m)$, and the signature formula assigns to each marked support a product of local kernel weights. The remaining distributional identities are therefore naturally studied through size-preserving bridges on compositions and their marked supports.

We encode L by its separator word

$$w(L) = 0^{\ell_1 - 1} 1\, 0^{\ell_2 - 1} 1 \cdots 1\, 0^{\ell_m - 1};$$

the ones mark the cuts between consecutive parts.

For a signature Σ and a host composition L, let $\mathrm{Occ}_\Sigma(L)$ denote the set of occurrences of Σ in L. For $k \geq 0$, a k-marked object is a pair (L, M) with $M \subseteq \mathrm{Occ}_\Sigma(L)$ and $|M| = k$.

Lemma 4.7 (Marked-occurrence principle)**.** *Suppose that, for every n and $k \geq 0$, there is a size-preserving bijection between the k-marked objects of size n for Σ and those for Σ'. Then Σ and Σ' are equidistributed on $\mathcal{L}$.*

Proof. For fixed n and k, put

$$B_{\Sigma,n,k} = \sum_{L\models n} \binom{P_\Sigma(L)}{k}.$$

This is exactly the number of k-marked objects of size n. Hence the assumed bijections give $B_{\Sigma,n,k} = B_{\Sigma',n,k}$ for every $k \geq 0$. Using $q^r = (1 + (q-1))^r = \sum_{k\geq 0} \binom{r}{k}(q-1)^k$, we obtain

$$\begin{aligned} D_{\Sigma,n}(q) &= \sum_{L\models n} q^{P_\Sigma(L)} \\ &= \sum_{k\geq 0} B_{\Sigma,n,k}(q-1)^k. \end{aligned}$$

The same expansion holds for Σ'. Since the binomial moments $B_{\Sigma,n,k}$ and $B_{\Sigma',n,k}$ agree for every k, the two distribution polynomials are equal for every n. $\square$

The residual identities are organized by three uniform bijective families, one exceptional Class 79 distributional identity, and one classical Mahonian bridge.

Proposition 4.8 (Uniform cut-shift bijections)**.** *For $a, b \in \{0, 1, 2\}$, let*

$$\Sigma_{a,b} = (111; (a, b, 2); \{0, 3\}), \qquad \Sigma'_{a,b} = (12; (b, a+1); \{2\}).$$

Then $\Sigma_{a,b}$ and $\Sigma'_{a,b}$ are equidistributed on $\mathcal{L}$.

Proof. For a host composition $L = (\ell_1, \dots, \ell_m)$, the signature formula gives

$$P_{\Sigma_{a,b}}(L) = \mathbf{1}_{\ell_m=1} K_{1,a}(\ell_1) \sum_{j=2}^{m-1} K_{1,b}(\ell_j).$$

Thus every host with positive occurrence number has the form $L = (x, \beta, 1)$, where β is a nonempty composition. Define $\Phi(x, \beta, 1) = (\beta, x+1)$. For the target signature,

$$P_{\Sigma'_{a,b}}(\beta, x+1) = K_{2,a+1}(x+1) \sum_{c\in\beta} K_{1,b}(c).$$

The kernel identity $K_{2,a+1}(x+1) = K_{1,a}(x)$ for $a = 0, 1, 2$ therefore gives $P_{\Sigma_{a,b}}(x, \beta, 1) = P_{\Sigma'_{a,b}}(\Phi(x, \beta, 1))$.

The map is size preserving, and its inverse on positive-occurrence hosts is $(\beta, y) \mapsto (y-1, \beta, 1)$ for $y \geq 2$. Hence, for every n and every positive occurrence number, the corresponding hosts are in bijection. Since the total number of compositions of n is the same on both sides, the zero-occurrence hosts also occur equally often. Thus the two signatures are equidistributed. $\square$

These give the nine nonexceptional Q-bridges.

Proposition 4.9 (The exceptional Class 79 bridge)**.** *The signatures*

$$\Sigma = (111; (1, 2, 1); \{0, 3\}), \qquad \Sigma' = (12; (2, 1); \{1\})$$

are equidistributed on $\mathcal{L}$.

Proof. For a composition $L = (\ell_1, \dots, \ell_m)$, the first statistic is $P_\Sigma(L) = \#\{j : 2 \le j \le m-1,\ \ell_j = 1\}$, the number of interior singleton parts. Put $A(t) = t/(1-t)$ and $I(t, q) = qt + t^2/(1-t)$. The first and last parts are arbitrary, while every interior part contributes a factor q precisely when it is a singleton. Hence

$$\begin{aligned} G_\Sigma(t, q) &:= \sum_L q^{P_\Sigma(L)} t^{|L|} \\ &= 1 + A(t) + \frac{A(t)^2}{1 - I(t, q)} \\ &= \frac{1 - qt}{1 - (1+q)t + (q-1)t^2}. \end{aligned}$$

For the second signature,

$$P_{\Sigma'}(L) = \sum_{i=1}^{m-1} \mathbf{1}_{\ell_i = 1} (\ell_{i+1} - 1).$$

Let H_0 be the generating function for a suffix when the preceding part is not a singleton, or when there is no preceding part, and let H_1 be the corresponding series when the preceding part is a singleton. Decomposing according to the next part gives

$$\begin{aligned} H_0 &= 1 + tH_1 + \frac{t^2}{1-t} H_0, \\ H_1 &= 1 + tH_1 + \frac{t^2 q}{1-tq} H_0. \end{aligned}$$

Solving these equations yields

$$H_0 = \frac{1 - qt}{1 - (1+q)t + (q-1)t^2} = G_\Sigma(t, q).$$

Since H_0 is the distribution series for Σ', the two signatures are equidistributed. □

This supplies the exceptional Q-bridge in Class 79.

Proposition 4.10 (Unified contraction bridges)**.** *For $z \in \{0, 1, 2, 3\}$, put*

$$\Sigma_z = (12; (2, z); \{1\}), \qquad \Sigma'_z = (3; z+1; \varnothing).$$

For each z, Σ_z and Σ'_z are equidistributed on $\mathcal{L}$.

Proof. Because the first local parameter is $2 = r_1 + 1$, the first supporting layer is rigid. Hence every occurrence of Σ_z is supported on an adjacent pair $(1, b)$ with $b \geq 2$. Two distinct such support pairs cannot overlap: the second member of a support pair has size at least two and therefore cannot be the singleton first member of another support pair.

Consider a k-marked object. Group the marks according to their supporting pairs and simultaneously contract every support pair carrying at least one mark by $(1, b) \mapsto b + 1$. The identity $K_{2,z}(b) = K_{3,z+1}(b + 1)$ shows that the local occurrence sets on the two sides have the same cardinality. For $z \leq 2$, both may be canonically indexed by the subsets $\binom{[b-z]}{2-z}$, and for $z = 3$ both local occurrence sets are singletons. We transfer the marked local occurrences under this identification.

The inverse is equally explicit. Every target part carrying at least one mark has size $b + 1 \geq 3$; replace it by $(1, b)$ and transfer its marked local occurrences back under the same identification. Unmarked parts and unmarked support pairs are left unchanged.

Thus we obtain a size-preserving bijection between the k-marked objects for Σ_z and Σ'_z for every k. The result follows from Lemma 4.7. □

These four bridges account for four of the remaining identifications.

Proposition 4.11 (Uniform separator-complement bridges)**.** *For $a \in \{0, 1, 2\}$, let*

$$\Sigma_a = (111; (a, 1, 1); \{0, 1\}), \qquad \Sigma'_a = (12; (a, 1); \{0\}).$$

For $L = (\ell_1, \ldots, \ell_m)$ with $m \geq 2$, one has

$$P_{\Sigma_a}(L) = K_{1,a}(\ell_1)(m - 2), \qquad P_{\Sigma'_a}(L) = K_{1,a}(\ell_1) \sum_{j=2}^{m} (\ell_j - 1).$$

If $m = 1$, both statistics are zero.

Proof. For $m \geq 2$, the signature formula gives the two displayed formulas in the statement; for $m = 1$, both statistics are zero. When $m \geq 2$, write $w(L) = 0^{\ell_1 - 1} 1 v$ and define $\Psi(w) = 0^{\ell_1 - 1} 1 \overline{v}$, where $\overline{v}$ is obtained from v by exchanging 0 and 1. For compositions with one part, let Ψ be the identity.

The map Ψ is a size-preserving involution and leaves ℓ_1 unchanged. The numbers of 1's and 0's in v are $m - 2$ and $\sum_{j=2}^{m} (\ell_j - 1)$, respectively. Complementation exchanges these two quantities, so $P_{\Sigma_a}(L) = P_{\Sigma'_a}(\Psi(L))$. Thus Ψ is a size-preserving bijection proving the desired equidistribution. □

These three bridges account for three of the remaining identifications.

Proposition 4.12 (Class 118 MacMahon bridge)**.** *The signatures $\Sigma = (12; 02; \varnothing)$ and $\Sigma' = (12; 11; \varnothing)$ are equidistributed on $\mathcal{L}$.*

Proof. For the separator word w of a composition $L = (\ell_1, \ldots, \ell_m)$, we prove

$$P_\Sigma(L) = \sum_{\substack{j\geq 2\\ \ell_j \geq 2}} (\ell_1 + \cdots + \ell_{j-1}) = \mathrm{maj}(w),$$

$$P_{\Sigma'}(L) = \sum_{j=2}^{m} (j-1)(\ell_j - 1) = \mathrm{inv}(w).$$

Write the separator word as

$$w = 0^{\ell_1 - 1} 1 0^{\ell_2 - 1} 1 \cdots 1\, 0^{\ell_m - 1}.$$

For every $j \geq 2$ with $\ell_j \geq 2$, the separator immediately preceding the jth part occurs at position $\ell_1 + \cdots + \ell_{j-1}$ and is followed by a zero. Hence w has a descent 10 precisely at these positions, and every descent arises in this way. This proves the first displayed identity.

On the other hand, each of the $\ell_j - 1$ zeros belonging to the jth part has exactly $j-1$ separator ones preceding it, which proves the second displayed identity. MacMahon's equidistribution of major index and inversion number on binary words of fixed content [15] therefore proves the result. □

Together these give $9+1+4+3+1 = 18$ bridges in the remaining step. Supplementary Table S1 inventories the five families. Comparing their endpoints with the coordinate components shows that every bridge is effective: each joins two components that have not previously been joined. Hence the number of components drops from 164 to 146.

5 The complete classification

Theorem 5.1 (Complete length-three classification)**.** *The mesh patterns of length three with layered underlying permutation form exactly* 146 *nonzero distribution classes on* $\mathcal{L}$*. All mesh patterns with underlying permutation* 231 *or* 312 *form one additional zero distribution class. Consequently, all* $6 \cdot 2^{16}$ *length-three mesh patterns form exactly* 147 *distribution classes on* $\mathcal{L}$*.*

Proof. For the upper bound, Theorems 2.1 and 3.1 reduce the four occurring underlying permutations first to 1,600 effective shadings and then to 644 signatures. The exact pointwise quotient of Theorem 3.4 gives 523 classes, and Proposition 4.1 gives 286 pointwise–*rc* orbits. The complete unique-support classification replaces 114 of these orbits by 28 classes, giving 200. The coordinate identities have 36 effective unions and give 164, as recorded in Supplementary Table S2. Finally, the eighteen all-size bridges of Section 4.4, inventoried in Supplementary Table S1, are effective and give 146. Every reduction used for this upper bound is therefore justified by an identity or equidistribution valid for every host size.

For the lower bound, choose one representative signature from each of these 146 components. Supplementary Proposition S1 evaluates the occurrence formula of Theorem 3.1 and proves that every two distinct representatives have different distribution polynomials at some size $n \leq 7$. A difference at one size precludes equidistribution, so the components cannot merge and there are at least 146 nonzero distribution classes. Every mesh pattern with layered underlying permutation occurs in its underlying permutation itself, since there are no unselected points, so none of these classes is the zero class. Finally, 231 and 312 cannot occur in a layered host, and all of their shadings give the one zero distribution class. □

The multiplicity $\nu(\Sigma)$ restores the 1,600 effective shadings, and the inactive-cell factor restores all raw mesh patterns.

6 Discussion

The classification separates three different quotients. Active-cell reduction identifies mesh diagrams that cannot be distinguished on layered hosts. The signature formula identifies statistics that agree on every individual host. Equidistribution is a further quotient and may arise from a nontrivial bijection of compositions even when the pointwise statistics are different.

The value of the signature is therefore operational rather than merely notational. It converts the geometry of a mesh diagram into a short weighted support polynomial. That polynomial exposes kernel degrees, coordinate symmetries, cut shifts, and separator-word statistics. In this way the classification is obtained from a fixed sequence of reductions rather than 146 unrelated calculations.

Length three is also the first place where a genuinely nonlocal word identity appears: Class 118 becomes the classical equidistribution of major index and inversion number on binary words. At larger lengths, the same structural pipeline remains available; Proposition 3.3 already gives $S_4 = 4{,}712$, and additional word statistics and overlapping marked blocks may appear.

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A Uniform layer-coordinate relabellings

All coordinate relabellings needed here are drawn from the following eleven uniform coordinate permutations. A tuple $U = (u_1, \ldots, u_m)$ denotes the map $U(\ell_1, \ldots, \ell_m) = (\ell_{u_1}, \ldots, \ell_{u_m})$. The notation $a{:}b$ denotes the consecutive string from a to b, in increasing or decreasing order as appropriate; in particular, $2{:}m = 2, 3, \ldots, m$. Thus $U_1 = (1, 3{:}m, 2)$ means

$$(\ell_1, \ell_2, \ell_3, \ldots, \ell_m) \longmapsto (\ell_1, \ell_3, \ldots, \ell_m, \ell_2).$$

Every local orbit joined in Supplementary Table S2 consists of signatures with three supporting coordinates. Their statistics vanish when $m < 3$, and for these two sizes we take every U_i to be the identity. The displayed coordinate words apply for $m \geq 3$. Each U_i is uniform in m, and Lemma A.1 verifies the identities required by Theorem 4.6.

U_1	$(1, 3{:}m, 2)$	U_2	$(2{:}m-1, 1, m)$
U_3	$(1, m, 2{:}m-1)$	U_4	$(m-1, m, 1{:}m-2)$
U_5	$(2, 1, 3{:}m)$	U_6	$(m-1, 1{:}m-2, m)$
U_7	$(2{:}m, 1)$	U_8	$(1, m, m-1{:}2)$
U_9	$(m-1{:}1, m)$	U_{10}	$(3{:}m, 1, 2)$
U_{11}	$(1{:}m-2, m, m-1)$		.

Lemma A.1 (Uniform coordinate identities)**.** *Every one of the* 43 *source–target connections listed explicitly in Supplementary Table S2 satisfies, for every m and every positive host composition L,*

$$P_\Sigma(L) = P_{\Sigma'}(U_i(L)),$$

with the orientation specified by that connection. Thus all these connections satisfy the hypothesis of Theorem 4.6.

Proof. All signatures involved have $r = (1, 1, 1)$. Put

$$b_0(x) = x, \qquad b_1(x) = 1, \qquad b_2(x) = \mathbf{1}_{x=1}.$$

For $\Sigma = (111; z_1 z_2 z_3; E)$, form the multiset

$$\mathcal{W}_m(\Sigma) = \biguplus_{\phi \in \Phi_E(m)} \big\{\!\!\big\{ (\phi(j), z_j) : z_j \neq 1 \big\}\!\!\big\}.$$

It records the labelled supporting coordinates after the factors $b_1 = 1$ have been deleted, retaining multiplicity when several supports produce the same reduced word. The signature formula becomes

$$P_\Sigma(L) = \sum_{W \in \mathcal{W}_m(\Sigma)} \prod_{(a,d) \in W} b_d(\ell_a).$$

If $U = (u_1, \ldots, u_m)$, let $U_* W$ replace every label (a, d) by (u_a, d). Consequently,

$$\mathcal{W}_m(\Sigma) = U_* \mathcal{W}_m(\Sigma') \tag{A.1}$$

implies the required identity for every choice of the positive variables $\ell_1, \ldots, \ell_m$.

For the connections in Supplementary Table S2, identity (A.1) is checked symbolically from the displayed coordinate words. If no b_1 factor is deleted, U_i gives a bijection between the two sets of admissible support triples. If one or two b_1 factors are deleted, the multiplicity of a reduced word is the number of ways to insert the deleted coordinates in the gaps prescribed by E. The conditions defining $\Phi_E(m)$ say only that a gap is open or closed, that one of its endpoints is fixed, or that two consecutive supports are adjacent. Substitution of the word U_i sends each such interval of insertion positions bijectively to the corresponding target interval and preserves its endpoints and multiplicity. This is an identity of sets whose endpoints are symbolic functions of m; no bounded value of m is substituted. The complete source–target list in Supplementary Table S2 therefore makes the verification finite and reproducible. The numbers of connections using $U_1, \ldots, U_{11}$ are, respectively, $6, 5, 5, 3, 2, 6, 4, 4, 4, 2, 2$, which sum to 43.

The example in Section 4.3 is the case with no deleted factor. Two examples show explicitly how multiplicities are handled. For $U_1 = (1, 3{:}m, 2)$,

$$P_{(111;012;\{0,1\})}(L) = \ell_1 \sum_{k=3}^{m} \mathbf{1}_{\ell_k=1} = P_{(111;021;\{0,2\})}(U_1(L)),$$

where one constant factor has been deleted. With two constant factors,

$$P_{(111;112;\{0\})}(L) = \sum_{k=3}^{m} (k-2)\mathbf{1}_{\ell_k=1} = P_{(111;121;\{2\})}(U_1(L)).$$

For each of the remaining rows of Supplementary Table S2, direct substitution of the indicated U_i gives the same support-insertion calculation, with a cyclic shift, an endpoint swap, or a reversal of the relevant interval. Since these operations are bijections of the full symbolic interval, this proves (A.1) uniformly in m. When $m < 3$, both statistics vanish, as noted above. □

Supplementary Table S2 also records the final-class and local-orbit labels of every oriented identity. Its 43 edges give 36 effective unions, reducing the 88 relevant local orbits to 52 components. With the 28 unique-support classes and 84 unaffected singleton multiple-support orbits, this gives $28 + 84 + 52 = 164$.